KAIZEN MATHS

# Lesson starter

**HCF, LCM and Prime Factorisation**

Algebra

Name _____

**S1**

HCF($9x^3$, $18x^4$)

**S2**

LCM($4x^2y$, $6x^4y^3$)

**S3**

HCF($9x^4y$, $15x^2y^3$)

**S4**

LCM($10x^3$, $4x^4$)



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# Exit ticket

**HCF, LCM and Prime Factorisation**

Algebra

Name _____

**E1**

HCF($9x^3$, $15x^2$)

**E2**

LCM($10x^3$, $12x^2$)

**E3**

HCF($6x^2y$, $9x^3y^3$)



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# Teacher answers & working

**HCF, LCM and Prime Factorisation · Algebra**

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## S1

$\text{HCF}(9x^3, 18x^4)$

**Answer:**

$9x^3$

**Working:**

Find the HCF of $9x^3$ and $18x^4$

$9x^3 = 9 \times x^3$

$18x^4 = 18 \times x^4$

HCF of coefficients: HCF(9, 18) = 9

Use the lowest shared powers: $x^3$

Combine the coefficient HCF with the lowest shared variable powers: $9 \times x^3$

$\text{HCF} = 9x^3$

## S2

$\text{LCM}(4x^2y, 6x^4y^3)$

**Answer:**

$12x^4y^3$

**Working:**

Find the LCM of $4x^2y$ and $6x^4y^3$

$4x^2y = 4 \times x^2 \times y$

$6x^4y^3 = 6 \times x^4 \times y^3$

LCM of coefficients: LCM(4, 6) = 12

Use the highest required powers: $x^4$, $y^3$

Combine the coefficient LCM with the highest required variable powers: $12 \times x^4 \times y^3$

$\text{LCM} = 12x^4y^3$

# Teacher answers & working

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## S3

$\text{HCF}(9x^4y, 15x^2y^3)$

**Answer:**

$3x^2y$

**Working:**

Find the HCF of $9x^4y$ and $15x^2y^3$

$9x^4y = 9 \times x^4 \times y$

$15x^2y^3 = 15 \times x^2 \times y^3$

HCF of coefficients: HCF(9, 15) = 3

Use the lowest shared powers: $x^2$, y

Combine the coefficient HCF with the lowest shared variable powers: $3 \times x^2 \times y$

$\text{HCF} = 3x^2y$

## S4

$\text{LCM}(10x^3, 4x^4)$

**Answer:**

$20x^4$

**Working:**

Find the LCM of $10x^3$ and $4x^4$

$10x^3 = 10 \times x^3$

$4x^4 = 4 \times x^4$

LCM of coefficients: LCM(10, 4) = 20

Use the highest required powers: $x^4$

Combine the coefficient LCM with the highest required variable powers: $20 \times x^4$

$\text{LCM} = 20x^4$

# Teacher answers & working

**HCF, LCM and Prime Factorisation · Algebra**

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## E1

$\text{HCF}(9x^3, 15x^2)$

**Answer:**

$3x^2$

**Working:**

Find the HCF of $9x^3$ and $15x^2$

$9x^3 = 9 \times x^3$

$15x^2 = 15 \times x^2$

HCF of coefficients: HCF(9, 15) = 3

Use the lowest shared powers: $x^2$

Combine the coefficient HCF with the lowest shared variable powers: $3 \times x^2$

$\text{HCF} = 3x^2$

## E2

$\text{LCM}(10x^3, 12x^2)$

**Answer:**

$60x^3$

**Working:**

Find the LCM of $10x^3$ and $12x^2$

$10x^3 = 10 \times x^3$

$12x^2 = 12 \times x^2$

LCM of coefficients: LCM(10, 12) = 60

Use the highest required powers: $x^3$

Combine the coefficient LCM with the highest required variable powers: $60 \times x^3$

$\text{LCM} = 60x^3$

# Teacher answers & working

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**E3**

$\text{HCF}(6x^2y, 9x^3y^3)$

**Answer:**

$3x^2y$

**Working:**

Find the HCF of $6x^2y$ and $9x^3y^3$

$6x^2y = 6 \times x^2 \times y$

$9x^3y^3 = 9 \times x^3 \times y^3$

HCF of coefficients: HCF(6, 9) = 3

Use the lowest shared powers: $x^2$, y

Combine the coefficient HCF with the lowest shared variable powers: $3 \times x^2 \times y$

$\text{HCF} = 3x^2y$